

Networking for AI side meeting

Discussion on charter for future Research Group

Research group on Networking for AI: Why?

Several BoFs, IETF WGs and side meetings are discussing topics associated with Net4AI:

- Discovery of Agents, Workloads, and Named entities (DAWN)
- Dynamic Multi-agent Secured Collaboration (DMSC)
- Agent Communication Protocols (AGENTPROTO)
- OAUTH / WIMSE...

⇒ Working group typically target short to medium term concerns and standards

⇒ A research group might:

- Target middle to long term issues
- Make a bridge with the research community

Positioning wrt. other BoFs and working groups

DAWN:

- Major focus on DNS-inspired discovery mechanisms, with uncertainty about scalability and trust
- ⇒ *Capability-based discovery, beyond DNS*

ACP:

- Reuse of real-time communication protocols and messaging
 - Incremental adaptations to address agents requirements
- ⇒ *Latent space communication*
- ⇒ *KV cache based communications*

Net4AI's goal is to have a clean slate discussion about what would be needed for distributed and multi-stakeholder AI

In the research world

The interest for networking aspects of AI system is growing in major conference venues.

- SIGCOMM'26: Roughly **45 - 55 of the 110 accepted papers** have a substantial AI angle
 - Training networking
 - Inference networking
 - LLMs used as networking tools
- OSDI'26: Roughly 60% of the papers in the technical program address AI infrastructure issues
 - LLM serving
 - Distributed AI
 - Memory systems

⇒ Growing interest for distributed AI systems

In the research world

... yet some topics are in a blind spot for now:

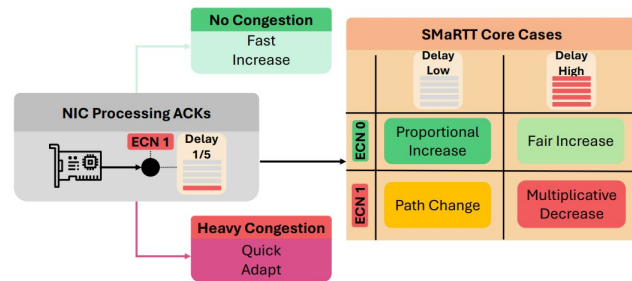
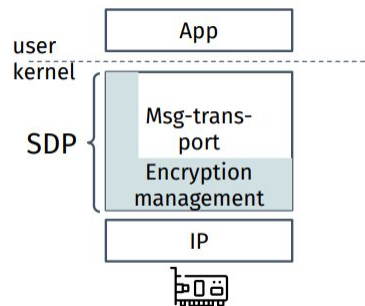
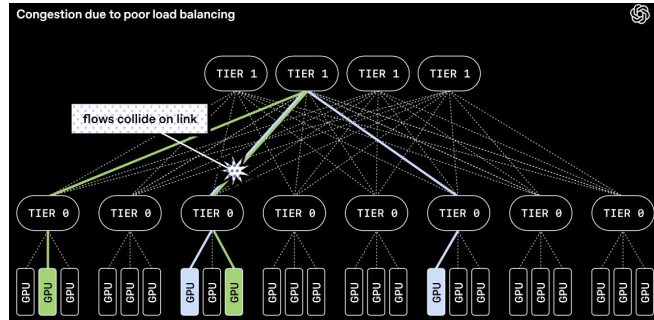
- Architectural Internet support for autonomous agents
- Principled decentralization
- Network architectures for agents
- Internet-native support for AI ecosystems

⇒ Research group objective: Gathering a community of researchers to discuss those topics and trigger more research

Possible topics for work in RG (1 / 4)

Usage and adaptation of internet protocols to operate AI workloads

- Multipath Reliable Connection
New RDMA-based transport protocol spraying traffic across many paths and actively load-balancing between them, eliminating the issue of flow collisions
- Transport-Level Encryption for Datacenter Networks
SDP integrates TLS-based encryption with a message-based transport protocol supporting efficient RPCs
- Sender-based Marked Rapidly-adapting Trimmed & Timed Transport



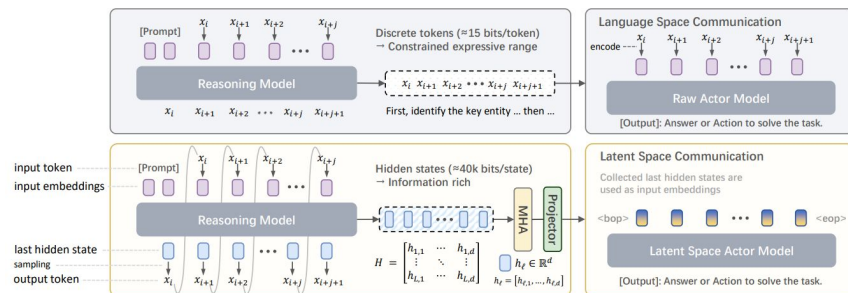
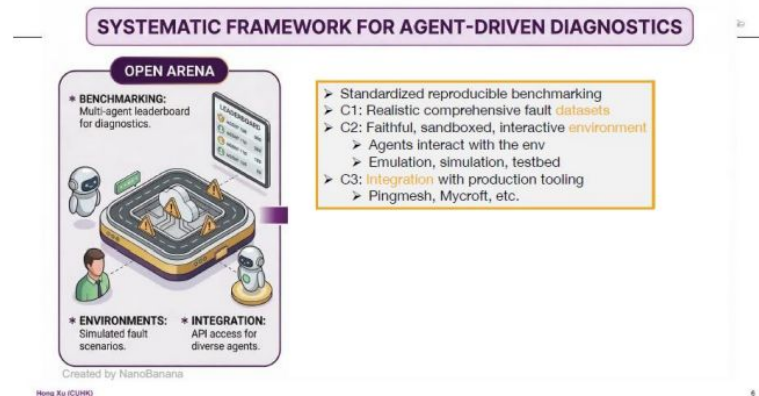
Possible topics for work in RG (2 / 4)

Research initiatives towards agentic management of networks require **testbed** and **benchmark** to foster idea exchange, and benchmark solutions between one another

- See Nika presentation as an example work to be presented

Agent-specific communication protocols

- Beyond anthropomorphic communications
- Latent space communication



Possible topics for work in RG (3 / 4)

Caching / content distribution for AI workloads

- KV-centric disaggregation of LLM inference engine first done inside a single datacenter realm, but paving the way for KV-centric networking at scale to distribute inference workloads beyond a DC's realm

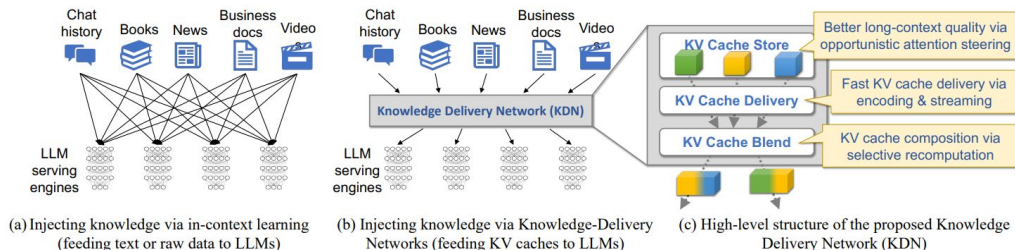
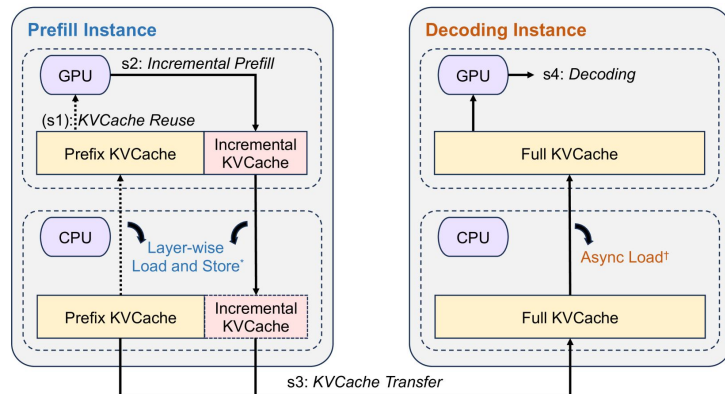


Figure 2. Architecture of a Knowledge Delivery Network (KDN).

Possible topics for work in RG (4 / 4)

Agent identity

- Principled discussion encompassing ***cross domain*** and ***decentralized identity***
- Get naming and addressing right in relationship with ***security as a foundation for trustworthy agent to agent communications***

New security requirements of Agentic AI

- Trust across domains for dynamic workloads
- Attestation

ADDRESSING AGENTIC AI NETWORKING CHALLENGES

Starting point: get naming right, then build security into agentic AI

- **DNS as a unifying namespace for cyberspace**
 - DNS: decentralized name management with TLD coordination and name delegation
 - DNS: offering *globally unique, semantically meaningful* names
 - Every entity — organization, user, agent, and service — have DNS names as primary identifiers
 - Semantic meaningfulness makes trust reasoning human-navigable
- **Identity = Name + Key**
 - Name provides semantic context; key provides cryptographic verifiability
 - Trust chains must be human-navigable; machine-verifiability alone is not adequate
- **Crypto protections anchor on local trust**
 - Global namespace, local trust
 - AI agents make this a scaling requirement, not an architecture option
 - Delegation chains are native infrastructure: scoped, verifiable, multi-hop

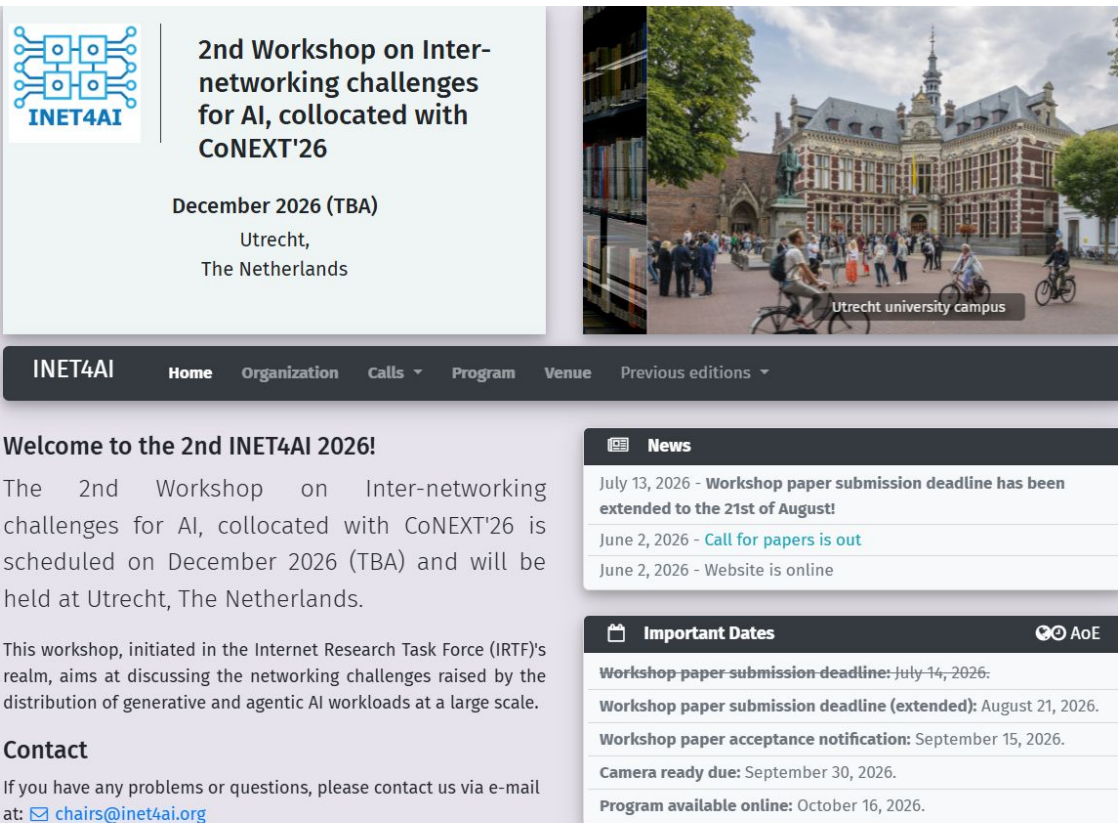
Potential Activities

- Assessing & developing experimental tools (simulators, testbeds)
- Collecting and sharing datasets for reproducible experiments
- Driving research on networked systems for AI (not limited to datacenters)
 - New architectures (KV-Cache-Centric, Mixture-of-Experts)
 - New requirements (inter-DC communication, security)
- Developing principled approaches on Agentic Communication (e.g., “Internet of Agents”)

INET4AI Workshop @ ACM CoNEXT 2026

- Call for paper open until **August 21st**

- Don't hesitate to submit your papers !



The screenshot shows the official website for the 2nd Workshop on Inter-networking challenges for AI, colloated with CoNEXT'26. The header features the INET4AI logo, the workshop title, the date (December 2026 TBA), and the location (Utrecht, The Netherlands). A navigation bar includes links for Home, Organization, Calls, Program, Venue, and Previous editions. The main content area has a 'Welcome to the 2nd INET4AI 2026!' section with a paragraph about the workshop's goals. To the right, there are two panels: 'News' with three items (deadline extension, call for papers, website online) and 'Important Dates' with four items (submission deadline, acceptance notification, camera ready due, program available online). A background image of the Utrecht university campus is visible on the right side of the header.

INET4AI | Home | Organization | Calls | Program | Venue | Previous editions

Welcome to the 2nd INET4AI 2026!

The 2nd Workshop on Inter-networking challenges for AI, colloated with CoNEXT'26 is scheduled on December 2026 (TBA) and will be held at Utrecht, The Netherlands.

This workshop, initiated in the Internet Research Task Force (IRTF)'s realm, aims at discussing the networking challenges raised by the distribution of generative and agentic AI workloads at a large scale.

Contact

If you have any problems or questions, please contact us via e-mail at: chairs@inet4ai.org

News

- July 13, 2026 - Workshop paper submission deadline has been extended to the 21st of August!
- June 2, 2026 - [Call for papers is out](#)
- June 2, 2026 - Website is online

Important Dates

- Workshop paper submission deadline:** July 14, 2026.
- Workshop paper submission deadline (extended):** August 21, 2026.
- Workshop paper acceptance notification:** September 15, 2026.
- Camera ready due:** September 30, 2026.
- Program available online:** October 16, 2026.